Types of Big data analytics

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Big Data Analytics: An Overview

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Introduction to Big Data Analytics

Big Data Analytics involves analyzing vast and complex datasets, which can be structured, semi-structured, or unstructured. It helps uncover patterns, trends, and correlations, finding applications across business, healthcare, finance, and social media.

Complex Datasets

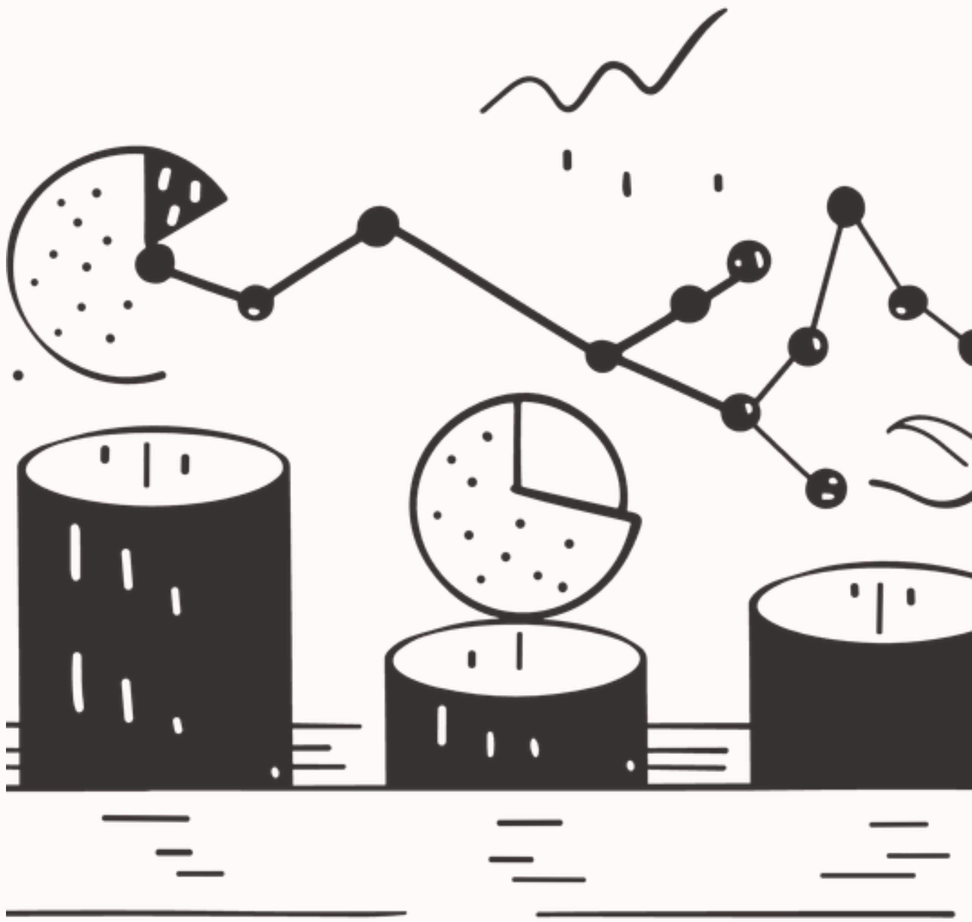
Analyzing large, diverse data.

Pattern Discovery

Uncovering trends and correlations.

Broad Applications

Used in many industries.



Why Big Data Analytics?

Traditional data processing tools often fall short when dealing with the scale and complexity of big data. Big Data Analytics provides the necessary capabilities to:

- Enable faster, better decision-making.
- Improve operational efficiency.
- Support data-driven strategies.

Faster Decisions

Improved Efficiency

Data-Driven Growth

Descriptive Analytics

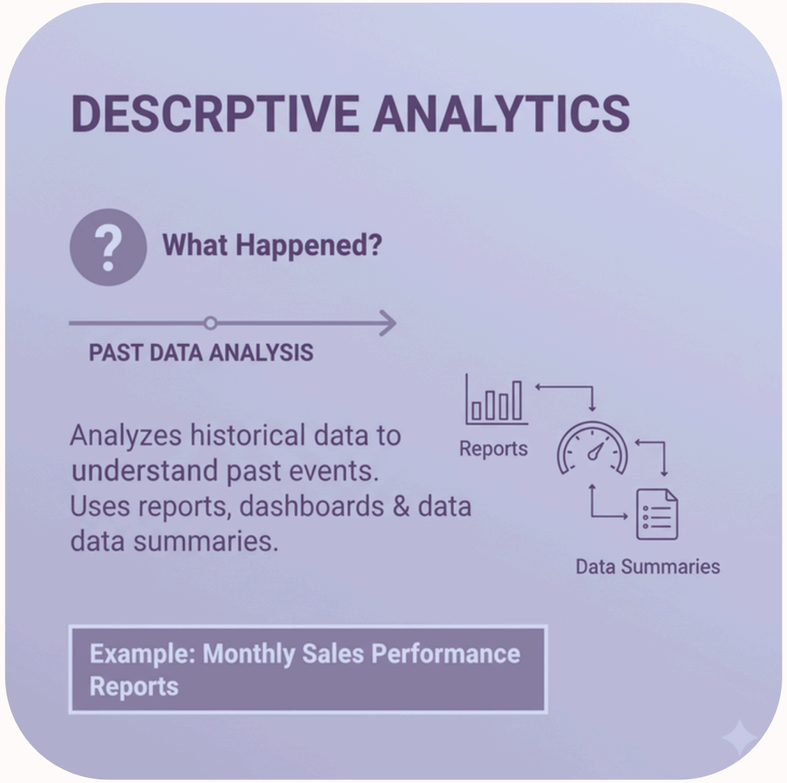
Descriptive analytics focuses on understanding past events by analyzing historical data. It answers the fundamental question: **What happened?**

- Uses reports, dashboards, and data summaries.
- Example: Monthly sales performance reports.

Past Data Analysis

"What Happened?"

Reports & Summaries



Diagnostic Analytics

Diagnostic analytics delves deeper into past outcomes to identify the root causes. It seeks to answer: **Why did it happen?**

- Employs data mining and drill-down analysis.
- Example: Investigating reasons for a sales decline.

Root Cause Analysis

"Why It Happened?"

Data Mining

Predictive Analytics

Predictive analytics uses historical data to forecast future events. It addresses the question: **What is likely to happen?**

- Leverages machine learning and statistical models.
- Examples: Demand forecasting, customer churn prediction.

Future Forecasting

"What Will Happen?"

Machine Learning

Prescriptive Analytics

Prescriptive analytics recommends actions to achieve desired outcomes. It answers: **What should be done?**

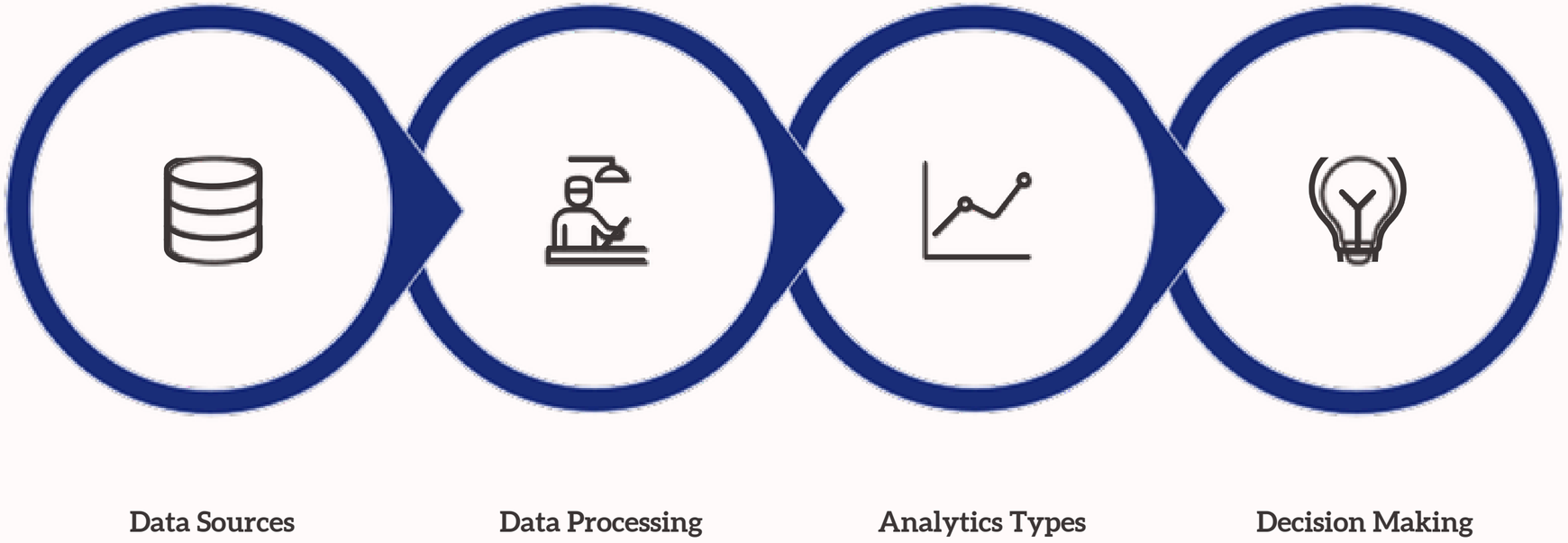
- Utilizes optimization and simulation techniques.
- Example: Recommendation systems for products or services.

Actionable Insights

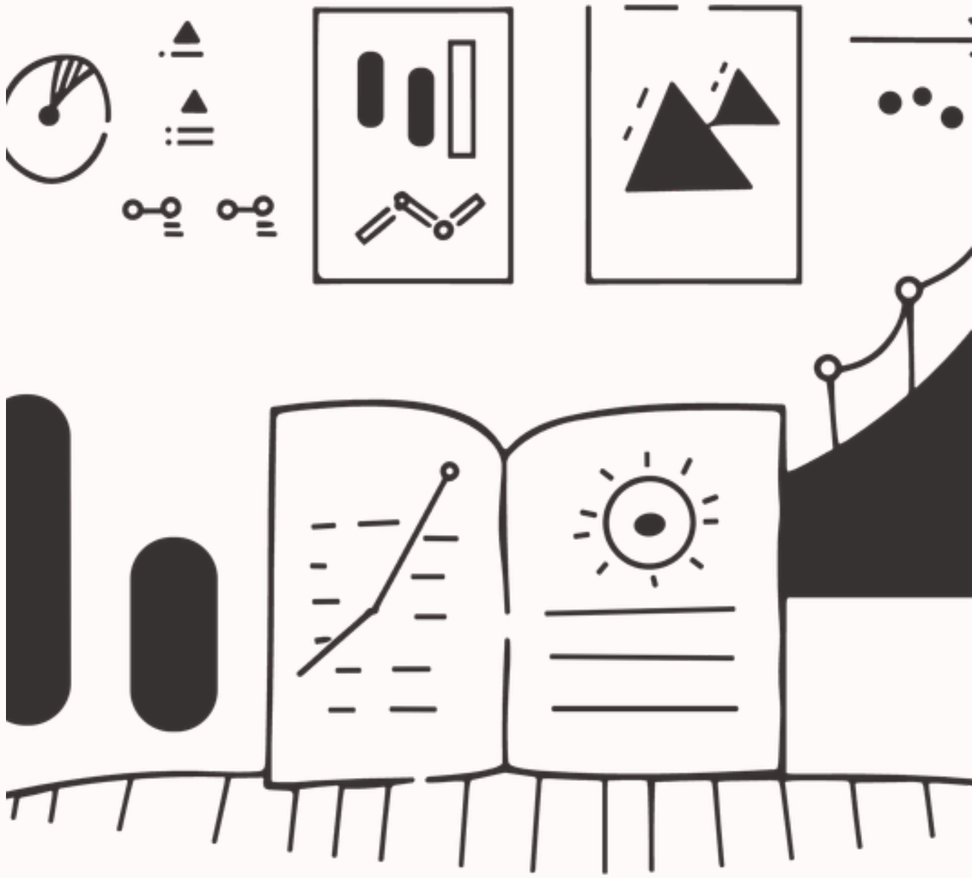
"What Should Be Done?"

Optimization

Block Diagram of Big Data Analytics



This diagram illustrates the flow of data through the Big Data Analytics pipeline, from raw sources to informed decision-making.

References

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